

SPSS应用

多元统计分析方法

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主要内容:

一、方差分析: GLM (general linear model)

析因设计、正交设计、重复测量设计、协方差分析

二、回归与相关分析: regression correlate

直线回归与相关分析、多元线性回归分析、
Logistic回归分析

三、生存分析: survival

四、其他多元统计分析方法:

主成分分析与因子分析、典型相关分析、
判别分析与聚类分析

方差分析: GLM (general linear model)

析因设计的方差分析

正交设计的方差分析

协方差分析

Analyze → general linear model → univariate: ...

重复测量设计的方差分析

Analyze → general linear model

→ repeated measures: ...

析因设计和正交设计的方差分析:

Analyze

→ general linear model

→ univariate

→ dependent: 应变变量

fixed factor: 组别变量

→ model: custom: 选择分析的因素 → continue

→ ok

析因设计: 例11-1 例11-2 例11-3

正交设计: 例11-4 例11-5



1 : druga

	druga	drug
1	1	
2	1	
3	1	
4	2	
5	2	
6	2	
7	3	
8	3	
9	3	
10	1	
11	1	
12	1	
13	2	
14	2	
15	2	
16	3	
17	3	
18	3	
19	1	

- Reports ▶
- Descriptive Statistics ▶
- Tables ▶
- Compare Means ▶
- General Linear Model ▶**
- Generalized Linear Models ▶
- Mixed Models ▶
- Correlate ▶
- Regression ▶
- Loglinear ▶
- Classify ▶
- Data Reduction ▶
- Scale ▶
- Nonparametric Tests ▶
- Time Series ▶
- Survival ▶
- Multiple Response ▶
- Missing Value Analysis...
- Complex Samples ▶
- Quality Control ▶
- ROC Curve...



- Univariate...**
- Multivariate...
- Repeated Measures...
- Variance Components...

2	120
2	100
3	75

Var

☐ Univariate

Dependent
镇痛时间 (min) [y]

Fixed Factor(s):
druga
drugb

Random Factor(s):

Covariate(s):

WLS

Model...
Contrasts...
Plots...
Post Hoc...
Save...
Options...

OK Paste Res

Univariate: Model

Specify Model
☐ Full factorial ☒ Custom

Factors & Covariates:
druga(F)
drugb(F)

Build Term(s)
Interaction

Model:
druga
drugb
druga*drugb

Sum of squares: Type III ☐ Include intercept in model

Continue Cancel Help

Tests of Between-Subjects Effects

Dependent Variable: 镇痛时间(min)

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	21466.667 ^a	8	2683.333	6.916	.000
Intercept	326700.000	1	326700.000	842.091	.000
druga	6572.222	2	3286.111	8.470	.003
drugb	7022.222	2	3511.111	9.050	.002
druga * drugb	7872.222	4	1968.056	5.073	.006
Error	6983.333	18	387.963		
Total	355150.000	27			
Corrected Total	28450.000	26			

a. R Squared = .755 (Adjusted R Squared = .645)

整个模型的检验

A因素、B因素及
A、B因素的交互作用

例11-4

例11-04.sav [DataSet4] - SPSS Data Editor

File Edit View Data Transform Analyze Graphs Utilities Window Help



20 :

	a	b	ab	c	d	x	var
1	1.00	1.00	1.00	1.00	1.00	86.00	
2	1.00	1.00	1.00	2.00	2.00	95.00	
3	1.00	2.00	2.00	1.00	2.00	91.00	
4	1.00	2.00	2.00	2.00	1.00	94.00	
5	2.00	1.00	2.00	1.00	2.00	91.00	
6	2.00	1.00	2.00	2.00	1.00	96.00	
7	2.00	2.00	1.00	1.00	1.00	83.00	
8	2.00	2.00	1.00	2.00	2.00	88.00	
9							
10							
11							

这个变量不需要

Univariate

ab

Dependent
x

Fixed Factor(s):
b
c

Random Factor(s):

Covariate(s):

WLS Weight:

Model...
Contrasts...
Plots...
Post Hoc...
Save...
Options...

OK Paste Reset Cancel Help

Univariate: Model

Specify Model
☐ Full factorial ☒ Custom

Factors & Covariates:
a(F)
b(F)
c(F)
d(F)

Build Term(s)
Interaction

Model:
a
b
c
d
a*b

Sum of squares: Type III

☒ Include intercept in model

Continue Cancel Help

Tests of Between-Subjects Effects

Dependent Variable: x

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	141.000 ^a	5	28.200	11.280	.083
Intercept	65522.000	1	65522.000	26208.800	.000
a	8.000	1	8.000	3.200	.216
b	18.000	1	18.000	7.200	.115
c	60.500	1	60.500	24.200	.039
d	4.500	1	4.500	1.800	.312
a * b	50.000	1	50.000	20.000	.047
Error	5.000	2	2.500		
Total	65668.000	8			
Corrected Total	146.000	7			

a. R Squared = .966 (Adjusted R Squared = .880)

**A因素、B因素、C因素、D因素
及A、B因素的交互作用**

重复测量设计资料的方差分析:

资料的输入格式: 参见例12-03

定义重复测量的
观察指标

Analyze → general linear model

→ repeated measures: **define factors:**

within-subject factor name : time (或factor1)

number of levels : 重复测量的次数

→ add → define

→ within subjects variables: t0 t1 t2 t3 t4

between subjects factors: group

→ model: custom: within subjects model: time

between subjects model: group → continue

→ ok

例12-3

File Edit View Data Transform Analyze Graphs Utilities Window Help									
1 : No 1									
	No	group				t2	t3	t4	
1	1								117
2	2								123
3	3								118
4	4								120
5	5								126
6	6					118	131	137	
7	7					119	129	133	
8	8					126	135	142	
9	9					111	123	131	
10	10					116	123	133	
11	11					118	135	129	
12	12					121	148	132	
13	13					120	143	136	
14	14					116	145	126	
15	15					118	142	130	
16									
17									
18									
19									
20									

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- Data Reduction
- Scale
- Nonparametric Tests
- Time Series
- Survival
- Multiple Response
- Missing Value Analysis...
- Complex Samples
- Quality Control
- ROC Curve...

- Univariate...
- Multivariate...
- Repeated Measures...
- Variance Components...

Repeated Measures Define Factor(s)

Within-Subject Factor Name:

Number of Levels:

Add

time (5)

Change

Remove

Measure Name:

Add

Change

Remove

Define

Reset

Cancel

Help

Repeated Measures

序号 [No]

Within-Subjects Variables
(time):

t0 (1)

t1 (2)

t2 (3)

t3 (4)

t4 (5)

Between-Subjects Factor(s):

组别 [group]

Covariates:

OK

Paste

Reset

Cancel

Help

Model...

Contrasts...

Plots...

Post Hoc...

Save...

Options...

Repeated Measures: Model



Specify Model

☐ Full factorial

☒ Custom

Within-Subjects:

time

Within-Subjects Model:

time

Build Terms



Between-Subjects:

group

Between-Subjects Model:

group

Interaction ▼

Sum of squares:

Type III ▼

Continue

Cancel

Help

Mauchly's Test of Sphericity

Measure: MEASURE 1

Within Subjects Effects	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon ^a		
					Greenhouse-Geisser	Huynh-Feldt	Lower-bound
time	.293	12.785	9	.178	.679	1.000	.250

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

b.

Design: Intercept+group

Within Subjects Design: time

**球对称检验：P>0.05
表明球对称**

Tests of Within-Subjects Effects

重复测量间即受试者内方差分析表

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.
time	Sphericity Assumed	2336.453	4	584.113	106.558	.000
	Greenhouse-Geisser	2336.453	2.715	860.644	106.558	.000
	Huynh-Feldt	2336.453	4.000	584.113	106.558	.000
	Lower-bound	2336.453	1.000	2336.453	106.558	.000
time * group	Sphericity Assumed	837.627	8	104.703	19.101	.000
	Greenhouse-Geisser	837.627	5.430	154.272	19.101	.000
	Huynh-Feldt	837.627	8.000	104.703	19.101	.000
	Lower-bound	837.627	2.000	418.813	19.101	.000
Error(time)	Sphericity Assumed	263.120	48	5.482		
	Greenhouse-Geisser	263.120	32.577	8.077		
	Huynh-Feldt	263.120	48.000	5.482		
	Lower-bound	263.120	12.000	21.927		

球对称检验： $P > 0.05$ 表明球对称所以看第一行结果。不对称时看下面校正的结果
该表为各时间点间差异和交互效应的检验

Tests of Between-Subjects Effects

组间即受试者间
方差分析表

Measure: MEASURE_1

Transformed Variable: Average

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Intercept	1155433.080	1	1155433.080	14649.223	.000
group	912.240	2	456.120	5.783	.017
Error	946.480	12	78.873		

各组别即诱导方法间的检验

协方差分析:

Analyze → general linear model → univariate:

dependent variable : 应变量 (即观察指标 y)

Fixed factors : 分组变量 (c)

表示交互作用，无统计学意义时才可以进一步分析，并把此项去除

Covariate : 协变量 (x)

→ model : custom : c, x, c*x → continue

→ options : display means for : c

compare main effects

→ continue

→ ok



	x	y	c	var	var	v
1	10.80	9.40	1.00			
2	11.60	9.70	1.00			
3	10.60	8.70	1.00			
4	9.00	7.20	1.00			
5	11.20	10.00	1.00			
6	9.90	8.50	1.00			
7	10.60	8.30	1.00			
8	10.40	8.10	1.00			
9	9.60	8.50	1.00			
10	10.50	9.10	1.00			
11	10.60	9.20	1.00			
12	9.90	8.40	1.00			
13	9.50	7.60	1.00			
14	9.70	7.90	1.00			
15	10.70	8.80	1.00			
16	9.20	7.40	1.00			
17	10.50	8.60	1.00			

General Linear Model ▶	GLM GEN	Univariate...
Generalized Linear Models ▶	GLM MULT	Multivariate...
Mixed Models ▶	GLM REP	Repeated Measures...
Correlate ▶		Variance Components...
Regression ▶		
Loglinear ▶	00	
Neural Networks ▶	00	
Classify ▶	00	
Dimension Reduction ▶	00	
Scale ▶	00	
Nonparametric Tests ▶	00	
Forecasting ▶	00	
Survival ▶	00	
Multiple Response ▶	00	
Missing Value Analysis... ▶	00	
Multiple Imputation ▶	00	
Complex Samples ▶	00	
Quality Control ▶	00	
ROC Curve... ▶	00	

- GLM GEN Univariate...
- GLM MULT Multivariate...
- GLM REP Repeated Measures...
- Variance Components...

[illegible]



Univariate



Dependent Variable:

3个月后 [y]



Fixed Factor(s):

组别 [c]



Random Factor(s):



Covariate(s):

入组前 [x]



WLS Weight:

Model...

Contrasts...

Plots...

Post Hoc...

Save...

Options...

Bootstrap...

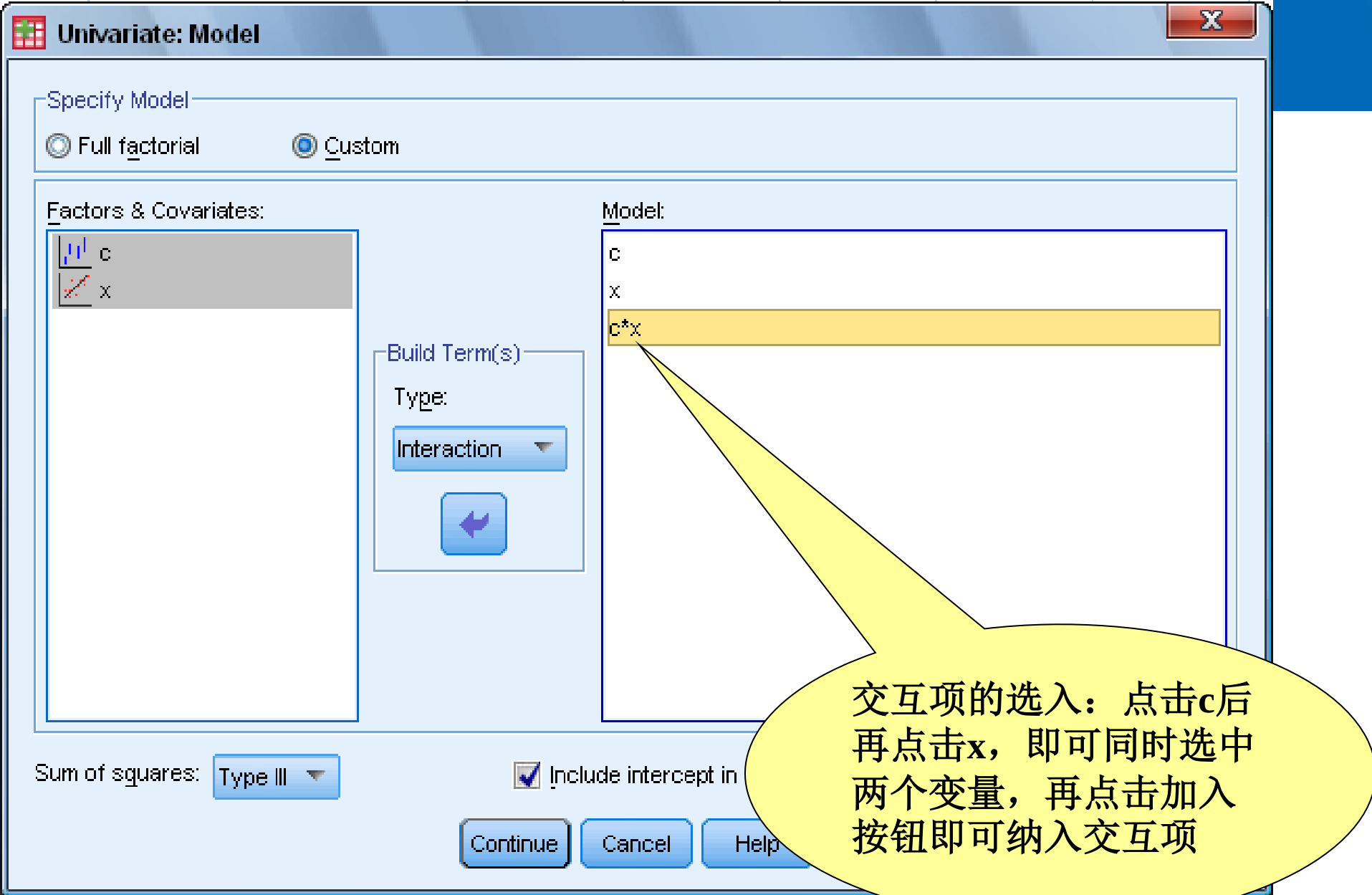
OK

Paste

Reset

Cancel

Help



Tests of Between-Subjects Effects

Dependent Variable: 3个月后糖化血红蛋白

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	48.973 ^a	5	9.795	56.618	.000
Intercept	.370	1	.370	2.141	.147
c	.202	2	.101	.585	.559
x	29.419	1	29.419	170.061	.000
c * x	.065	2	.033	.188	.829
Error	14.531	84	.173		
Total	6293.520	90			
Corrected Total	63.504	89			

a. R Squared = .771 (Adjusted R Squared = .758)

交互项无统计学意义，可把此项从模型中去除后进一步分析。

Tests of Between-Subjects Effects

Dependent Variable: 3个月后糖化血红蛋白

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	48.908 ^a	3	16.303	96.053	.000
Intercept	.363	1	.363	2.140	.147
c	19.851	2	9.925	58.480	.000
x	30.183	1	30.183	177.835	.000
Error	14.596	86	.170		
Total	6293.520	90			
Corrected Total	63.504	89			

a. R Squared = .770 (Adjusted R Squared = .762)

修正均数:

Estimates

Dependent Variable: 3个月后糖化血红蛋白

组别	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
试验组	8.356 ^a	.076	8.206	8.506
盐酸二甲双胍	8.877 ^a	.075	8.727	9.027
试验药+盐酸二甲双胍	7.728 ^a	.075	7.578	7.877

a. Covariates appearing in the model are evaluated at the following values:
入组前糖化血红蛋白 = 9.9700.

两两比较:

Pairwise Comparisons

Dependent Variable: 3个月后糖化血红蛋白

(I) 组别 (J) 组别		Mean Difference (I-J)	Std. Error	Sig. ^a	95% Confidence Interval for Difference ^a	
					Lower Bound	Upper Bound
试验组	盐酸二甲双胍	-.521 [*]	.107	.000	-.734	-.308
	试验药+盐酸二甲双胍	.628 [*]	.107	.000	.416	.840
盐酸二甲双胍	试验组	.521 [*]	.107	.000	.308	.734
	试验药+盐酸二甲双胍	1.149 [*]	.106	.000	.938	1.361
试验药+盐酸二甲双胍	试验组	-.628 [*]	.107	.000	-.840	-.416
	盐酸二甲双胍	-1.149 [*]	.106	.000	-1.361	-.938

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

a. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

Thank You